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Assignment 6 – Views (Module 6)

Introduction

Most of what a SQL database does is hold data in tables, but the people using the database usually aren't writing SELECT statements from scratch. They're running queries somebody saved ahead of time. There are three main ways to save that code: views, functions, and stored procedures. The questions below look at when a view is the right pick and where it fits relative to the other two.

Explain when you would use a SQL View.

A view is basically a saved SELECT. If I find myself running the same query a lot, or if it's something other people will need to run without learning the joins themselves, I'd save it as a view and let everyone call it by name. The other big reason to use views is access control. When a table has sensitive columns, I can deny direct access to the table and put a view in front of it that shows only what I want exposed. That keeps the underlying table protected while still giving people the data they need.

Explain the differences and similarities between a View, Function, and Stored Procedure.

All three save reusable SQL inside the database under a name you can call later, which is the main thing they share. The differences start with how flexible each one is. A view holds a single SELECT and can't accept parameters. A function can accept parameters and return either one value or a full table, so it can shape its result based on input. A stored procedure is more flexible still. It can run any kind of SQL, take parameters, return multiple result sets, and change data through INSERT, UPDATE, or DELETE. The Module 6 notes recommend sticking with views when you can and only stepping up to a function or procedure when the view can't do what you need.

Conclusion

The three tools overlap but each one is suited to a different kind of job. Views are simple and best for read-only reporting. Functions are useful when results need to change based on input. Stored procedures are the catch-all, especially anywhere data must be modified. Pick the simplest one that still does what you need and don't reach for something more complicated unless you have a reason to.